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First Semester QUESTION BANK

Problem Solving Techniques

Blueprint

Section	MODULE 1	MODULE 2	MODULE 3	MODULE 4	MODULE 5
	Number of Questions	Number of Questions	Number of Questions	Number of Questions	Number of Questions
A (5 marks)	2	1		1	1
B (9 marks)		1	1	0.5	0.5
C (12 marks)			0.33	0.33	0.33
Total Marks	10	14	13	14	13

Note: Module wise questions for the blueprint can be taken in any combination like

Section A – 1, 1, 1,1, 1 or 2, 1,1,1 or 2,2,1 or 2,3

SECTION – A					
1. 5 Marks Questions (1:6 Ratio)					
		Blooms Level	CO	Module	Marks
1.	Discuss the use of flowchart with a suitable example.	L2	1	1	5
2.	Explain top-Down approach for problem solving with an example.	L2	1	1	5
3.	Discuss differences between Top-Down and Bottom-Up approach in an algorithm design.	L2	1	1	5
4.	Describe the steps involved in solving a problem.	L2	1	1	5
5.	Explain Bottom-Up approach for problem solving with an example.	L2	1	1	5
6.	Write the pseudo code to find the maximum number in a list of 5 integers.	L6	2	1	5
7.	Illustrate a flowchart to check whether a given number is even or odd.	L3	2	1	5
8.	Define an algorithm and describe its characteristics.	L2	1	1	5
9.	Discuss the concept of an algorithm with a suitable example.	L2	1	1	5
10.	Explain various data types in C used to represent data in memory.	L2	1	1	5
11.	Illustrate basic symbols involved in a flowchart with example	L3	1	1	5

12.	Write an algorithm to calculate simple interest for a given principal amount, rate and year.	L6	2	1	5
13.	What are preprocessor directives in C? Explain any two with an example.	L2	3	2	5
14.	Explain ternary and logical operators in C with examples.	L2	2	2	5
15.	Write a program to check if the given number is prime or not.	L6	2, 4	2	5
16.	Describe the rules and syntax of writing a valid switch statement. Mention when it is preferred over if-else.	L2	3	2	5
17.	Illustrate the differences between while, do-while, and for loops in terms of syntax and execution.	L3	3	2	5
18.	Write a program to display factorial of a given number	L6	4	2	5
19.	Define pointers. Write a program to declare and initialize the pointer variable.	L6	3,4	4	5
20.	Define recursion. Write a recursive function in C to calculate the factorial of a number.	L6	3,4	4	5
21.	Write a C function to find the sum occurrence of a digit in a number.	L6	3,4	4	5
22.	Explain various methods to define user defined functions with syntax and examples.	L2	3,4	4	5
23.	Illustrate how to use argc and argv to read three integers from the command line and print their sum.	L2	3,4	4	5
24.	Write a C program to swap two numbers using pointers. Explain how * and & are used.	L6	3,4	4	5
25.	Define Structure and write the general syntax for declaring and accessing members of a structure.	L2	3	5	5
26.	Differentiate between structure and union using suitable examples.	L4	3,4	5	5
27.	Explain Enumerated data type and typedef. Give its syntax and example.	L2	3	5	5
28.	Write a C program to read and display the information of a student in a class using structures.	L6	4	5	5
29.	Write a C program to print name, USN, subject and IA marks of a student using union.	L6	4	5	5
30.	Define a structure type Book with book name, author, pages and price. Write a program to read the data and display the same.	L6	4	5	5
31.	Evaluate the following preprocessor code snippet and discuss its potential impact on the program. #define PI 3.14159 #undef PI #define PI 3.14 int main() {	L5	5	2	5

	<pre>printf("Value of PI: %f\n", PI); return 0; }</pre>				
32.	Differentiate between compile-time errors and run-time errors. How can debugging tools assist in resolving run-time errors?	L4	5	5	5
33.	Explain any two commonly used debugging tools and their key features.	L2	5	5	5
SECTION - B 2. 9 Marks Questions (1:6 Ratio)					
		Blooms Level	CO	Module	Marks
1.	i) Write a C program to display the multiplication table of 1 to n numbers entered by the user.	L6	4	2	5
	ii) Discuss the syntax of for loop along with an example.	L2	3		4
2.	Discuss any four input and output functions in C with syntax and example.	L2	3	2	9
3.	i) Implement a menu driven C program to perform addition, subtraction, multiplication and division of given two numbers based on choice.	L6	4	2	6
	ii) Differentiate between an if-else ladder and a switch statement with suitable examples.	L4	3		3
4.	Write a C program to read 6 subjects marks and calculate the total and percentage and display the respective grade. Note: Refer the below percentage for the grade - Percentage $\geq 90 \rightarrow$ Grade A - Percentage $\geq 75 \rightarrow$ Grade B - Percentage $\geq 50 \rightarrow$ Grade C - Otherwise $\rightarrow$ Fail	L6	3, 4	2	9
5.	i) Write a C program to check if the given number is palindrome or not.	L6	3, 4	2	5
	ii) Explain how the use of do-while can simplify input validation problems. Provide a program example where the loop ensures at least one valid input.	L2	3		4
6.	i) Trace the execution of the following C program and determine the final output: <pre>#include <stdio.h> int main() { int p = 3, q = 5, result; result = (++p > q--) ? p + q : p - q; printf("p = %d, q = %d, result = %d\n", p, q, result); return 0; }</pre>	L4	5	2	5

	} ii) Describe the role of relational and bit-wise operators with an example.	L2	3		4
7.	Write a C program to perform following operation without using built-in functions: i) concatenate two strings ii) copy two strings iii) find length of the given string	L6	3,4	3	9
8.	i) Write a C program to read N numbers into an array and perform Linear search. ii) Define array. Explain the declaration and initialization of a 2D array.	L6 L2	3,4	3	4 5
9.	Write a program in C to input a string and perform the following operations using user-defined functions: i) Find the length of the string ii) Reverse the string iii) Compare two strings	L6	3,4	3	9
10.	Write a C program to multiply two matrices of different order.	L6	3,4	3	9
11.	i) Describe how Bubble sort works and sort the list of numbers using bubble sort 46,32,23,78,12,86,17. ii) Write a C program to find the sum of array elements using a recursive function.	L2 L6	3,4	3	4 5
12.	i) Write a C program to read the elements in an array and print the same in reverse order. ii) Define string. List out all string manipulation functions. Explain any two with example	L6 L2	3,4	3	5 4
13.	i) Explain how different types of arrays are declared and initialized in C. ii) Write a C program to find the largest element given in an array of elements.	L2 L6	3, 4	3	5 4
14.	i) Write a structure to store the name, account number and balance of customers. Print the details of the customer based on the given account number. ii) Explain call by value and call by reference with an example program each.	L6 L2	3, 4	5 4	4 5
15.	i) Compare and contrast malloc(), calloc(), and realloc() functions. Also explain the use of free() with suitable examples. ii) Write a program using a structure pointer to store and print book details (title, author, price).	L4 L6	3, 4	4 5	5 4
16.	i) Write a C program to print the values stored in the dynamically allocated array.	L6	3, 4	4	5

	ii) Explain the use of structure pointers in C. How do you declare and use a pointer to a structure?	L2		5	4
17.	i) Differentiate Structure, Union and Array with suitable examples.	L4	3, 4	5	4
	ii) Write a C program to display the fibonacci series up to n using recursion.	L6		4	5
18.	i) Explain the concept of nested structures in C programming with the help of a suitable example.	L2	3, 4	5	4
	ii) Write a program for swapping two numbers without using the third variable by using Call by Value and Call by Reference method.	L6		4	5
19.	i) Explain with an example how recursion can be replaced with iteration.	L2	3, 4	4	5
	ii) Write a C program to compare the size of structure and union having the same members (int, float, char) and explain the difference	L6		5	4
	SECTION – C				
	3. Case study or application-based question for 12 marks (1:6 Ratio)				
		Blooms Level	CO	Module	Marks
1.	i) What are the advantages of passing structures to functions using pointers instead of by value?	L2	3,4	5	4
	ii) Write a user defined function to sort the array elements by passing array as parameter to the function and returning array from the function.	L6		4	4
	iii) Write a C function to find the sum occurrence of a digit in a number.	L6		3	4
2.	i) Write a C program to demonstrate the use of nested structure by storing and displaying a student's details along with address (street, city, pin code).	L6	3,4	5	4
	ii) Discuss any 4 library functions using suitable examples.	L2		4	4
	iii) Write a C program to find the maximum and minimum values in an array.	L6		3	4
3.	i) Describe the purpose of a debugger, and how does it help in program development?	L2	3,4	5	4
	ii) Write a C program that uses pointers to perform basic arithmetic operations.	L6		4	4
	iii) Write a C program that reads a sentence and prints the frequency of each of the vowels and total count of consonants?	L6		3	4
4.	i) What is the output of the following "C" programs? (Show the step by step tracing output) int main() {	L4	3,4, 5	5	4

	<pre> int a = 33 , b =17; while (a!= b) if (a > b) a-=b; else b-=a; printf("%d\n", a); return 0; } </pre>				
	ii) Why is modular programming important? How do functions help in achieving it?	L2		4	4
	iii) Write a C program to pass an array to a function and calculate the average of its elements.	L6		3	4
5.	i) What is the output of the following "C" programs? (Show the step by step tracing output) <pre> int main() { int i = 0 , j =0; do { if (i%3 ==0) { ++j; printf("%d\n", j); } ++i; } while(i < 15); return 0; } </pre>	L4	3,4, 5	5	4
	ii) Explain the general syntax of a function definition, declaration, and call in C	L2		4	4
	iii) Write a C program to display the transpose of a given matrix.	L6		3	4
6.	i) Write a C program to illustrate passing a union variable to a function and printing its value.	L6	3,4	5	4
	ii) Write a C program that uses functions to calculate the area and perimeter of a rectangle. (functions with parameter and return value)	L6		4	4
	iii) Illustrate the strlen(), strcat(), strcpy() and strcmp() function with suitable examples.	L3		3	4